
Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

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Abstract

1 Cross-skeleton motion generation trains generative models to carry action structure
2 and motion intention from one body to another. Yet an action-consistent target
3 motion admits two observationally compatible explanations: source-preserving
4 transfer and target-action recovery. We show that this ambiguity is structural
5 rather than incidental: under standard generative objectives, the source-conditioned
6 retargeting map is non-identifiable in sparse heterogeneous motion domains. Un-
7 paired marginal matching yields *gauge non-identifiability*: a relative gauge between
8 skeleton-specific latent spaces allows different source-conditioned maps to induce
9 the same training evidence. Sparse paired supervision admits the complementary
10 failure mode, *conditional-mean degeneration*: ambiguous action-level pairings
11 drive squared-error objectives toward a target-action prototype that is independent
12 of the source clip’s motion intention. To make the missing evidence observable,
13 we introduce *Source-Instance Fidelity* (SIF), a diagnostic that tests whether the
14 source-side geometry remains visible in the generated target motions after the
15 target skeleton and action are fixed. Under this diagnostic, standard action-level
16 success across animal motion domains recurrently coincides with behavior at the
17 *source-blind floor*, while positive cases localize source-instance information to
18 auxiliary motion-space constraints that narrow the observational equivalence class.
19 The implication is that retargeting requires objectives and evaluations capable of
20 identifying the source-conditioned map that a retargeting claim asserts. Project
21 page: <https://keen-semolina-09811d.netlify.app/>.

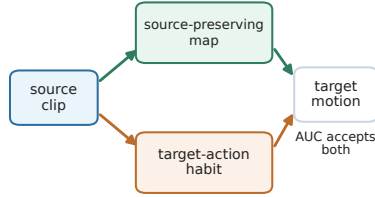
1 Introduction

23 Motion generation is increasingly expected to work across bodies rather than within a single fixed
24 skeleton. This expectation has grown from retargeting models that encode skeletal structure to
25 generative and correspondence-based models designed for heterogeneous bodies [Aberman et al.,
26 2020, Lee et al., 2023, Wang et al., 2024, Gat et al., 2025, Chen et al., 2025b]. A motion clip may
27 come from a human, a quadruped, or an articulated animal, and a target body may have different
28 proportions, joint topology, and kinematic limits. Cross-skeleton retargeting is the problem of
29 generating a motion for the target body that keeps the action and the motion intention of the source
30 clip, not merely a visually plausible motion on the target skeleton.

31 Most empirical evidence has been read through cross-skeleton contrastive retrieval under the standard
32 protocol introduced with large heterogeneous motion libraries [Gat et al., 2025, Truebones Motions,
33 2025], measured by the area under the receiver-operating-characteristic curve (AUC), which sum-
34 marizes a discrimination task rather than a retargeting claim. The generated target motion is treated
35 as a query: target-library clips with the requested action are positives, clips with other actions are
36 negatives, and the metric estimates the probability that a positive clip ranks above a randomly selected

A. Same score, different maps

Action recovery does not determine which map produced the target motion.



B. The missing axis

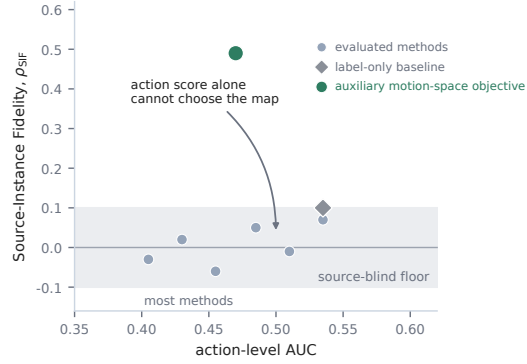


Figure 1: Action-level success and source-instance transport are different claims. The same target action can arise from a source-preserving map or from target-action recovery. Standard action-level evidence accepts both explanations, whereas an instance-level diagnostic asks whether changing the source clip changes the target motion in the corresponding way.

negative. This makes AUC an appropriate test of action recoverability in the target domain. It is silent, however, about whether the particular source clip selected the target motion that was produced.

The unsupported inference is the source-conditioned map. Retargeting claims that a particular source clip is mapped to a particular target motion, preserving timing and motion intention through the change of body. Yet an action-consistent output has two observationally compatible explanations: a source-preserving map may have transported the source instance, or a target-action habit may have produced a plausible motion for the requested body and action. Figure 1 summarizes this ambiguity. The action-level score accepts both explanations, while the missing source-instance question asks whether changing the source clip would change the target output accordingly.

This ambiguity becomes structural in sparse heterogeneous motion libraries. In the animal motion data we study, the Truebones zoo [Truebones Motions, 2025], motions span bodies with different skeletal topologies and joint counts: the overwhelming majority of body-and-action combinations are unobserved, and many of the observed ones contain only a single clip. Such data provide action labels and per-skeleton target examples, but not dense source-target correspondences that specify how a given source instance should move on a new body. The map asserted by retargeting is therefore not selected by the evidence that standard generative objectives consume.

The resulting limit is an identifiability limit, in the sense that the observed training evidence can be compatible with multiple latent explanations [Locatello et al., 2019, Khemakhem et al., 2020]. Under standard objectives in sparse heterogeneous domains, the source-conditioned retargeting map is *non-identifiable*. Per-skeleton marginal matching admits a relative gauge between skeleton-specific latent spaces, so distinct source-to-target maps can produce the same training evidence, a *gauge non-identifiability*. In the paired case, cell-paired squared-error regression under random target-cell pairing selects the target-action prototype rather than the source-conditioned motion, a *conditional-mean degeneration*. These mechanisms explain why better action recovery does not necessarily identify the retargeting map.

The missing observable is source-instance fidelity rather than target-action recognizability. We introduce *Source-Instance Fidelity* (SIF), a diagnostic that tests whether the source-side geometry remains visible in the generated target motions after the target skeleton and action are fixed. In controlled synthetic data, where the true correspondence is known, SIF separates source-preserving transport from source-blind generation. In sparse animal motion data, it reveals that action-level success can coexist with behavior near the *source-blind floor*.

Reading cross-skeleton generation through this lens reorganizes what counts as evidence. Action recovery and source-conditioned transfer are no longer collapsed into a single quantity, the two structural mechanisms above expose why standard objectives cannot select between them, and SIF supplies the source-instance signal that the action-level view never asks for. The same reading isolates a characterized positive case: ACE [Li et al., 2023], an adversarial cross-embodiment model

that adds a motion-space constraint beyond per-skeleton distribution matching, ties its outputs back to source-side structure tightly enough to narrow the observational equivalence class left open by standard objectives.

The implication is that cross-skeleton retargeting cannot be evaluated only by asking whether the target action is recognizable. A generated motion may be plausible, action-consistent, and retrievable while still failing to express the motion intention of the source clip. What must be identified is the source-conditioned map itself, and the central scientific question is which evidence and inductive bias are sufficient to select it.

2 Related Work

Cross-skeleton motion retargeting. The retargeting literature has progressively weakened the assumption that source and target bodies share a convenient representation. Early neural retargeting made skeletal structure explicit, while later skeleton-agnostic representations and context-conditioned sequence models sought motion spaces that remain meaningful across body graphs [Aberman et al., 2020, Lee et al., 2023, Wang et al., 2024]. More recent work carries the same ambition into sparse heterogeneous libraries, where generative models and sparse-correspondence procedures must operate without dense paired examples [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b]. Our concern is orthogonal to this modeling trajectory: once an output has the requested action, the remaining question is whether the evidence identifies the particular source-conditioned map that retargeting claims.

Motion generation beyond fixed bodies. Recent motion generation work also shows that the fixed human skeleton is no longer the only natural domain of synthesis. The shared pressure across this broader literature is to represent motion when body identity, morphology, and topology become part of the conditioning problem rather than fixed background structure [Raab et al., 2024, Lee et al., 2025, Egan et al., 2024, Chen et al., 2025a]. These settings make body variation a central modeling object. The structural question here is complementary: when data are sparse and correspondences are not instance-level, action labels and target examples may specify what action should appear on the target body without specifying which source motion intention selected it.

Identifiability and distributional transport. Our formal argument is closest in spirit to identifiability limits in representation learning, where marginal fit alone does not recover latent structure without suitable inductive bias or auxiliary variables [Locatello et al., 2019, Khemakhem et al., 2020]. Distributional transport gives a second comparison point, since Schrödinger bridge objectives can learn a coupling between marginals without observing instance-level correspondences [Bortoli et al., 2021, Kim et al., 2024]. In cross-skeleton retargeting, however, the missing object is not merely a transport between two marginal motion distributions. It is the relative alignment that tells a target decoder how to interpret the source instance.

Motion libraries and retrieval comparators. Motion-library methods preserve realized target-side motion snippets by construction, and this property connects classical motion graphs to modern sparse-correspondence transfer [Kovar et al., 2002, Chen et al., 2025b]. This makes retrieval useful as a diagnostic foil: it can expose what action-level evidence accepts without claiming to synthesize a new source-conditioned motion. We use this distinction later to define a target-library comparator whose role is diagnostic rather than generative. Appendix A gives additional context for these connections.

3 The Source-Conditioned Map

The object at stake is not a target motion distribution, but a map conditioned on a particular source instance. Let $\mathcal{X}_s$ denote the motion space of skeleton s , with clips annotated by an action label c ; this notation records that clips on different bodies occupy different kinematic domains. A retargeting claim asserts a family of maps $T_{a \rightarrow b} : \mathcal{X}_a \rightarrow \mathcal{X}_b$ such that a source clip x_a on skeleton a is carried to skeleton b while preserving the action, timing, and motion intention of that clip. This is a stronger object than a conditional generator for plausible motions on b given the label c : the map must change when the source instance changes, even when the action and target skeleton are held fixed.

122 Sparse heterogeneous motion libraries provide much weaker evidence than this map requires. In the
 123 Truebones zoo used in our evaluation [Truebones Motions, 2025, Gat et al., 2025], the dataset contains
 124 70 skeletons and 616 clips across 90 exact actions, yielding 6,300 possible skeleton-action cells.
 125 Only 413 of these cells are occupied, a fill rate of 6.6%, and the median occupied cell contains one
 126 clip. Thus a learner often sees that a skeleton can perform an action, but not multiple within-action
 127 instances from which the geometry of motion intention could be inferred, much less a dense set of
 128 source-target correspondences specifying how the same instance should appear on a different body.

129 This sparsity matters because source-instance evidence is relational. To ask whether a map preserves
 130 the source instance, one must be able to compare several source clips that share a source skeleton and
 131 action, generate the corresponding motions on the same target skeleton, and then ask whether the
 132 source-side geometry remains visible after retargeting. Truebones contains only 49 source-target-
 133 action triples with at least three source clips each; later paired comparisons are necessarily restricted
 134 further to the 37 triples that all evaluated methods can support. The structural question is therefore
 135 not whether action labels are useful, since they plainly are, but whether this sparse evidence selects a
 136 source-conditioned map rather than a target-action habit.

137 The distinction fixes the burden for the rest of the paper. Action-label evidence constrains which
 138 target motions are admissible for a requested action, whereas source-instance evidence constrains how
 139 differences among same-action source clips should reappear after the body changes. The former can
 140 support recognizable cross-skeleton generation; only the latter can identify the map that a retargeting
 141 claim asserts. Section 4 makes this gap precise by showing how standard objectives can leave multiple
 142 source-conditioned maps consistent with the same observed evidence.

143 4 Structural Non-Identifiability

144 A model can make plausible target motions while the objective remains indifferent to the source-
 145 conditioned map that produced them. This section makes the previous ambiguity structural by
 146 isolating two objective-level mechanisms: per-skeleton marginal matching leaves a relative latent
 147 coordinate freedom between bodies, while cell-paired squared-error regression under random target-
 148 cell pairing selects a target-action prototype. Both mechanisms preserve the evidence consumed by
 149 the objective while changing, or erasing, the source-instance dependence that retargeting requires.

150 The first mechanism appears whenever skeleton-specific encoders and decoders are coupled only
 151 through per-skeleton latent marginals. Let P_s be the motion distribution for skeleton s , let $\mu_s =$
 152 $(E_s)_{\#}P_s$ be the latent marginal, and write a shared-latent objective schematically as

$$L(E, D) = \sum_s L_s(D_s \circ E_s; P_s) + \Omega(\mu_1, \dots, \mu_K), \quad (1)$$

153 where Ω depends only on the per-skeleton latent marginals. For measurable invertible latent transfor-
 154 mations $g = (g_1, \dots, g_K)$ with measurable inverses, define

$$G_{\Omega}(\mu) = \{g : \Omega((g_1)_{\#}\mu_1, \dots, (g_K)_{\#}\mu_K) = \Omega(\mu_1, \dots, \mu_K)\}. \quad (2)$$

155 For $g \in G_{\Omega}(\mu)$, set $E'_s = g_s \circ E_s$ and $D'_s = D_s \circ g_s^{-1}$. The relative transformations $g_b^{-1} \circ g_a$ are
 156 invisible to the marginal objective, but they are exactly the transformations that determine how a
 157 source latent is interpreted by a target decoder.

158 **Theorem 1** (Gauge non-identifiability). *Let L be the objective in Eq. (1) with Ω depending only*
 159 *on per-skeleton latent marginals, and let $g \in G_{\Omega}(\mu)$ as in Eq. (2). Then $L(E', D') = L(E, D)$.*
 160 *If, in addition, there exist a, b such that $D_b \circ g_b^{-1} \circ g_a \neq D_b$ on a set of positive μ_a -measure, the*
 161 *induced retargeting maps differ: $T'_{a \rightarrow b} = D_b \circ g_b^{-1} \circ g_a \circ E_a \neq T_{a \rightarrow b} = D_b \circ E_a$ on a set of positive*
 162 *P_a -measure. $T_{a \rightarrow b}$ is determined by L only up to the relative gauge $g_b^{-1} \circ g_a$.*

163 This theorem is narrower than a generic impossibility result for disentanglement [Locatello et al.,
 164 2019] or nonlinear independent component analysis. It does not say that no auxiliary information can
 165 identify a representation. It says that the objective class above, when supplied with sparse skeleton-
 166 and-action evidence, does not select a relative latent alignment between skeletons. Identifiability under
 167 iVAE-style auxiliary variables [Khemakhem et al., 2020] requires the auxiliary distribution to induce
 168 full-rank variation in the prior sufficient statistics; on Truebones, where only 413 skeleton-action
 169 cells out of 6,300 are occupied and the median occupied cell contains one clip, skeleton-and-action
 170 metadata do not provide the empirical support needed to invoke that condition.

For Gaussian-aligned regularizers, the invariance contains $O(d)^K$; with $K = 70$ skeletons and the $d = 32$ latent dimension used for the finite-dimensional lower-bound calculation, quotienting the diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, still leaves a lower bound of 34,224 continuous gauge degrees of freedom. This is not a count of distinct realized maps, because an observable retargeting ambiguity requires *decoder non-degeneracy*: the decoder must discriminate at least some directions on the gauge orbit. That condition is architecture-dependent. The checks in Appendix D record a small Kullback-Leibler (KL)-regularized VAE whose decoder nearly absorbs the orbit, with a gauge-to-noise ratio of 1.11, and a separate AnyTop hidden-state sweep showing that a transformer diffusion decoder for arbitrary skeleton topology [Gat et al., 2025] can make an orthogonal perturbation orbit visible after decoding. The AnyTop sweep is not the source of the $d = 32$ count; it is a decoder-observability boundary check. Orbit observability is nevertheless distinct from gauge-element selection: a decoder can distinguish rotation directions without the objective selecting which rotation aligns source and target latents. The main point is therefore not that every latent coordinate change is observable, but that the objective does not identify the relative latent alignment needed for a source-preserving map.

The second mechanism arises when sparse paired evidence is treated as if it were instance-level supervision. Consider a predictor trained with squared error from a source clip x_a and target skeleton b to a target motion x_b . If the target is drawn uniformly from the target skeleton and source-action cell rather than paired to the same motion instance, the risk is

$$L_{\text{paired}}(f) = \mathbb{E} [\|f(x_a, b) - x_b\|_2^2]. \quad (3)$$

The Bayes-optimal squared-error predictor is then the conditional mean of the target cell [Hastie et al., 2009]; in this regime, the conditional mean is a target-action prototype rather than a source-conditioned motion.

Proposition 1.1 (Conditional-mean degeneration). *Assume x_b is square-integrable, f ranges over measurable functions, and the empirical pairing draws targets uniformly at random from the (skeleton, action) cell $B(b, c) = \{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$ with $c = \text{action}(x_a)$. Then the (almost surely unique) minimiser of L_{paired} is $f^*(x_a, b) = |B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b$, the empirical cell mean: a single fixed prototype per (target skeleton, source action), independent of the specific source clip.*

The proposition is deliberately scoped. It applies to cell-paired squared-error regression under random target-cell pairing, not to true instance-level pairs, which would supply exactly the correspondence the sparse regime lacks. It also assumes the target cell is nonempty and that motions are represented in the Euclidean vector space on which the squared-error loss is evaluated. The controlled ladder in Figure 2 shows the same mechanism in a setting where the correct source-conditioned transport T^* is known; the oracle variance is the within-cell variance of $T^*(x_a)$ over held-out source clips. As paired support per cell ranges from $M = 1$ to $M = 50$, the trained squared-error regressor preserves only 13% to 25% of that oracle within-cell output variance. No evaluated Truebones method exactly instantiates this random cell-pairing model. The closest real-data echo is AL-Flow, a label-conditional flow comparator that receives cluster and exact-action labels but no source motion. The latent-level diagnostic reported in Section 5 places AL-Flow at -0.001 , indicating that source-instance information can be lost before decoding when the training signal approximates cell-level pairing rather than instance-level correspondence. Appendix C contains the formal proofs: the gauge theorem in Appendix C.1, the conditional-mean proposition in Appendix C.2, and the retrieval corollary in Appendix C.3.

These two mechanisms also explain why retrieval comparators can be useful diagnostics without identifying the retargeting map. The following formal check records a boundary condition: a rule whose ranking depends only on features computed from joint positions, rather than from a learned shared latent, is outside the gauge variables of the objective above. It can therefore characterize the source-blind floor without selecting the source-conditioned map.

Corollary 1.1 (Retrieval is gauge-invariant). *Let $r : X_a \times \{b\} \rightarrow L_b$ be a retrieval rule that maps a source to a clip in the target-skeleton library by a function of a feature ϕ that is itself gauge-invariant, for example kinematic descriptors computed directly from joint positions rather than from a learned latent. Then r is invariant under the gauge group G_Ω of any shared-latent objective: replacing E_a with $g_a \circ E_a$ leaves $r(x_a, b)$ unchanged. Furthermore $r(x_a, b)$ is a single existing clip rather than an average, so the conditional-mean operator of Prop. 1.1 does not apply.*

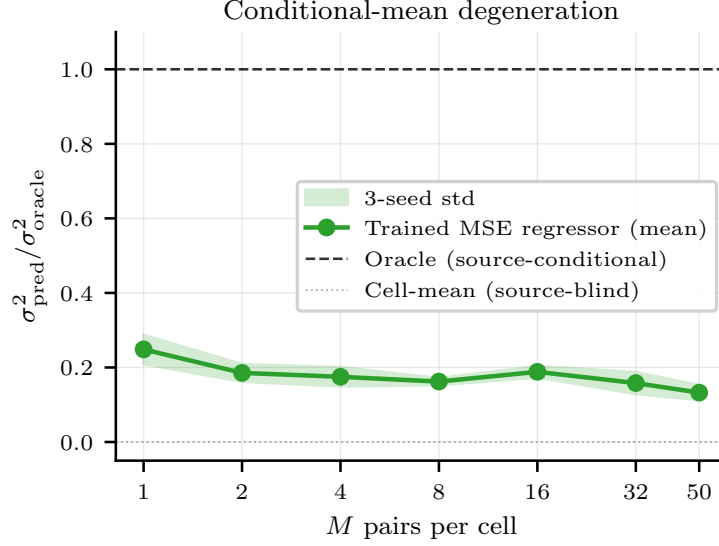


Figure 2: Cell-paired squared-error regression collapses toward the target-cell prototype. Across $M = 1$ to $M = 50$ pairs per cell in a controlled problem with known source-conditioned transport T^* , predicted output variance remains only 13% to 25% of the oracle within-cell variance $\text{Var}(T^*(x_a))$.

225 The corollary is a measurement statement, not a constructive recommendation, and its invariance
 226 follows from the fact that the score is computed in motion space rather than through the learned latent
 227 coordinates. We later instantiate this measurement role through label-only baselines and ANCHOR,
 228 our deterministic target-library retrieval comparator. These comparators can mark what action-level
 229 evidence accepts at the source-blind floor precisely because they avoid the two objective mechanisms
 230 analyzed here. They do not identify the retargeting map; they expose why standard evidence alone
 231 cannot.

232 5 Source-Instance Fidelity and Empirical Evidence

233 The missing observable is whether source variation remains ordered after retargeting. A source-
 234 conditioned map does not only produce a plausible target motion; it should preserve the relative
 235 geometry among source clips when the source skeleton, target skeleton, and action are fixed. We
 236 therefore define *Source-Instance Fidelity* (SIF) on a triple (a, b, c) with at least three distinct source
 237 clips. For source clips $x_i, x_j \in \mathcal{X}_a$ and generated target motions $y_i, y_j \in \mathcal{X}_b$, each pair is first
 238 cropped to its common duration and put into a common comparison frame by least-squares Procrustes
 239 superposition, which removes global translation, rotation, and isotropic scale before measuring
 240 joint-trajectory discrepancy [Goodall, 1991]. Let $d_{\text{src}}(i, j)$ be this aligned per-frame distance between
 241 x_i and x_j , and let $d_{\text{out}}(i, j)$ be the corresponding distance between y_i and y_j . Using Pearson’s
 242 product-moment correlation [Pearson, 1896], SIF is defined as

$$\rho_{\text{SIF}} = \text{CORR}(\{d_{\text{src}}(i, j)\}_{i < j}, \{d_{\text{out}}(i, j)\}_{i < j}), \quad (4)$$

243 computed within each eligible source-target-action triple. The reported per-method value is the macro-
 244 average of these per-triple correlations, with confidence intervals obtained by bootstrap resampling
 245 over triples.

246 For the failure-mode analysis, we also use *latent Source-Instance Fidelity* (L-SIF), the same relational
 247 statistic computed before decoding. The source distances remain $d_{\text{src}}(i, j)$, but the target-side
 248 distances are replaced by Frobenius distances between the corresponding target latents, mean-pooled
 249 over time or token index when sequence lengths differ [Golub and Van Loan, 1996]. L-SIF serves a
 250 complementary purpose: it identifies whether source-instance information is absent before decoding
 251 or whether it is present in the latent representation but erased by the decoder.

SIF asks a different question from action retrieval. If a generator recovers the right target action by returning a target-action habit, then changing the source clip need not change the target output in the corresponding way; the action-level score may be positive while ρ_{SIF} remains near zero. Conversely, a source-preserving map should make larger source-side changes appear as larger target-side changes, producing a positive correlation after the target skeleton and action have been fixed. The diagnostic is therefore deliberately relational: it tests the map asserted by retargeting rather than the recognizability of a single generated clip.

The synthetic calibration fixes the scale of the diagnostic. The controlled 2×2 setting is a synthetic motion environment in which the source-conditioned transport T^* is known for every source clip, providing the instance-level correspondence that the real corpus lacks. As visualized in Figure 13 in Appendix F, the oracle output $T^*(x_a)$ gives $\rho_{\text{SIF}} = +0.9957$ with a 95% confidence interval $[+0.9714, +1.0000]$. Two source-blind references sit at zero: sampling a random clip from the correct target cell gives $\rho_{\text{SIF}} = -0.0002$, and sampling random Gaussian motion gives $\rho_{\text{SIF}} = -0.0009$. Their upper bootstrap bounds are $+0.1382$ and $+0.1286$, respectively, defining an empirical source-blind floor around $+0.14$; the later shorthand $|\rho_{\text{SIF}}| < 0.10$ is a conservative way to describe behavior well inside this floor.

On the real corpus, SIF must be reported with the support that the sparse data allow. The largest Truebones SIF evaluation set contains 49 eligible triples, while the common support across all evaluated rows contains 37 triples. The comparison is organized by evidentiary role rather than by method ranking. Some rows ask whether published generative objectives preserve source-instance information when evaluated under the same target request [Gat et al., 2025, Kim et al., 2025, Li et al., 2023]; others ask how much of the evidence can be attributed to sparse correspondence, supervision-matched labels, latent transport, target-library retrieval, or label-only sampling [Chen et al., 2025b, Bortoli et al., 2021, Kim et al., 2024]. Appendix B records the source-conditioning adaptations needed for rows whose original form does not expose a source-input slot, and Appendix K records the scope of the DPG-SB-v3 row. For ACE and MoReFlow, the suffixes T and I denote transductive variants trained with all skeletons and inductive variants trained on the training skeleton split, respectively. We use each row’s largest available support for its primary estimate, because that gives the most stable confidence interval, and reserve the 37-triple intersection for paired comparisons that require identical triples across rows.

ANCHOR occupies a distinct role in this comparison: it is deliberately not a retargeting generator. Given a source clip, target skeleton, and action request, it predicts a coarse source action cluster from motion-space descriptors and retrieves an existing clip from the target library using that prediction, exact action metadata, and a kinematic descriptor computed directly from joint trajectories. This makes it useful as a comparator outside the learned latent objectives analyzed in Section 4: if such a rule attains action-level evidence but stays near the SIF floor, then action recovery and source-conditioned transfer are empirically separated. Appendix G gives the scoring rule. Its role is therefore diagnostic, not generative.

The main empirical pattern is that action-consistent generation usually does not preserve source-instance geometry. Figure 3 shows the largest support each evaluated row admits: seven learned rows, ANCHOR, and the two random label-only references all fall inside $|\rho_{\text{SIF}}| < 0.10$ on their max-support estimates, giving ten of fourteen evaluated rows. On the 37-triple common support, the support shift raises random-same-exact-action to a label-only positive ($\rho = +0.186$) rather than a learned source-preserving exception, while ACE-I is robust under clustered bootstrap and ACE-T separates from the floor under iid, source-skeleton, and target-skeleton bootstrap but is sensitive to action clustering. MoReFlow-T remains borderline, with lower confidence bounds of -0.003 on the 37-triple support and $+0.018$ on max support, so we do not treat it as a stable exception. The result is the empirical counterpart of the structural claim in Section 4: action-level evidence can be present while the source-conditioned map remains unidentified. Appendix E reports the corresponding support, bootstrap, and latent-SIF checks.

The qualitative source-substitution view in Appendix J makes the same measurement visible. The row labels use the two diagnostics introduced above: SIF describes the output geometry, while L-SIF locates whether the corresponding source geometry is present before decoding. The test does not ask whether each target motion looks like an attack; it asks whether different source attacks induce correspondingly different target attacks on the same target body. This is why SIF is the natural observable for the paper’s identifiability question: it turns the otherwise hidden source-conditioned

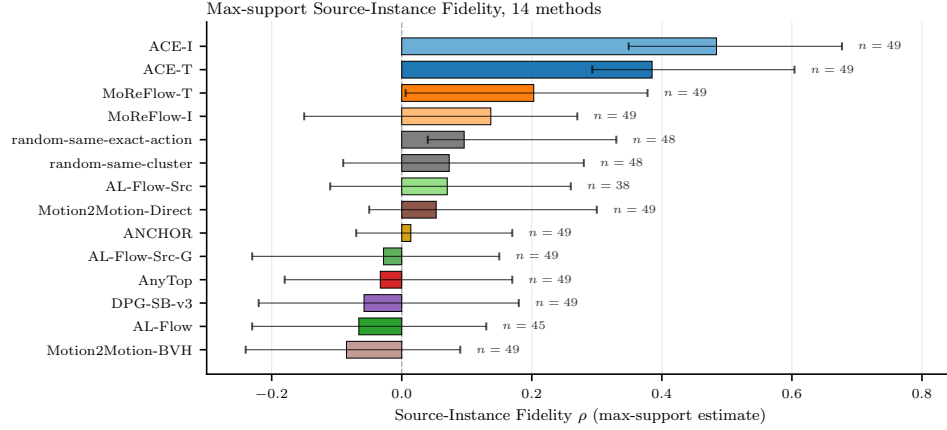


Figure 3: Source-Instance Fidelity on each evaluated row’s largest available Truebones support. The point is the macro-average SIF over eligible source-target-action triples, the interval is a bootstrap confidence interval over triples, and the printed n is the number of triples supporting that estimate. On max support ($n = 38$ to 49), most evaluated rows remain at the source-blind floor. ACE-I and ACE-T are the nontrivial positive rows highlighted by the SIF axis; label-only rows are interpreted by the action-level audit below.

map into a measurable source-substitution test without requiring dense source-target pairs in the training data.

The standard score says yes whenever the target action is recoverable, even when SIF says the source-conditioned map has not been selected. Table 1 gives the numerical audit, and Figure 4 visualizes the same comparison. The random-same-cluster reference chooses a random target clip with the same coarse action cluster, while random-same-exact-action chooses a random target clip with the exact action label whenever such a clip exists. These rules have no access to a source-conditioned retargeting map, yet random-same-cluster matches ANCHOR within ± 0.005 on both the overall and held-out splits. The exact-action variant exceeds ANCHOR by $+0.119$ overall and $+0.250$ held-out, but only on the 221/600 overall and 57/200 held-out coverage where exact-action positives exist in the target library. The qualifier is part of the claim: strong action-level evidence can be generated by label availability alone, without identifying the map. Appendix H gives the broader enumeration and action-level comparison tables behind this audit.

Table 1: Label-only retrieval can match or exceed ANCHOR’s action-level AUC without solving the retargeting problem. AUC is measured under the same cluster-tier Procrustes protocol used for ANCHOR. The exact-action reference has restricted coverage, so its larger gaps are only defined on the rows where the target library contains an exact-action candidate.

Method	Overall $n/600$	Held-out $n/200$	Overall AUC	Held-out AUC
ANCHOR	600	200	0.764	0.673
random-same-cluster	570	177	0.769 (+0.005)	0.678 (+0.005)
random-same-exact-action	221	57	0.883 (+0.119)	0.923 (+0.250)

Parentetical Δ values are method AUC minus ANCHOR. random-same-cluster matches ANCHOR within ± 0.005 on both splits. random-same-exact-action exceeds ANCHOR by $+0.119$ overall and $+0.250$ held-out, but only on the 221/600 and 57/200 coverage where exact-action positives exist; the coverage qualifier travels with the gap.

The label-only audit changes how the positive SIF rows should be read. Once action-level AUC can be reproduced by references with no source-conditioned map, the question is not whether a method also scores well on action recovery, but whether it contains independent motion-space evidence that preserves the source instance after the target skeleton and action are fixed. On the 37-triple cross-method intersection, where every evaluated method is scored on the same triples, ACE-T and ACE-I are the retargeting rows that remain above the source-blind floor; paired tests further distinguish ACE-I from both the random-same-exact-action reference and MoReFlow-T, the strongest nontrivial

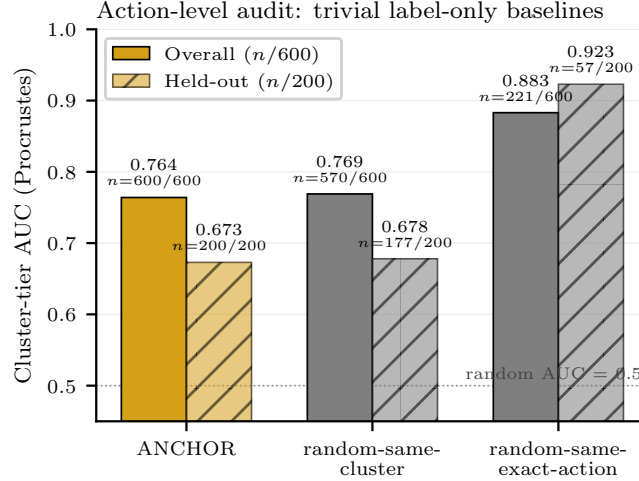


Figure 4: Action-level AUC cannot distinguish source-conditioned transport from label-based retrieval. ANCHOR retrieves from the target library using action information and motion descriptors, while the two random references use only label constraints. Their AUC values remain in the same action-level regime, showing that the standard score can accept target-action recovery even when source-instance transfer has not been established. The exact-action reference is coverage-limited to 221/600 overall and 57/200 held-out queries.

alternative. Thus ACE is not a counterexample to the action-level critique. It is the characterized case in which an auxiliary motion-space constraint supplies evidence outside the standard score.

The ACE controls in Appendix I localize that evidence to the motion-space adversarial loss rather than to source copying. Table 19 and Figure 18 show that removing this loss and retraining the variant across three seeds lowers SIF to a mean of 0.214, with range [0.175, 0.276], below the ACE baselines and close to the source-blind regime. The nearest-neighbor leakage check in Table 20 gives the complementary control: 76 to 79 percent of ACE outputs are closer, under coarse kinematic features, to a target-skeleton training clip than to the source clip, which does not support a direct source-copying explanation. The structural lesson is therefore specific: auxiliary motion-space objectives can help select a source-preserving element from the equivalence class, while action-level AUC by itself cannot certify that selection.

6 Conclusion

Cross-skeleton retargeting can appear successful under action-level evaluation while the source-conditioned map remains unidentified. The analysis isolates two structural mechanisms behind this ambiguity. *Gauge non-identifiability* leaves per-skeleton latent spaces related by an unresolved gauge under per-skeleton marginal matching, while *conditional-mean degeneration* makes cell-paired squared-error regression under random target-cell pairing recover a target-action prototype rather than the motion intention of the source clip. Source-Instance Fidelity then makes the missing evidence measurable by asking whether source-side geometry remains visible after target skeleton and action are fixed.

The empirical picture follows the structural analysis. In sparse heterogeneous animal motion data, action-consistent outputs often remain at the source-blind floor, while label-only references show why action-level AUC can stay positive without selecting a map. ACE is the characterized positive case in our study, because its auxiliary motion-space constraint preserves enough source-instance evidence to narrow the observational equivalence class. The broader implication is constructive: progress in retargeting should be judged by whether an objective improves action recovery, source-conditioned transport, or both. The open scientific question is which auxiliary motion-space objectives select a source-preserving map from the observational equivalence class identified here.

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437 A Extended Related Work

438 Cross-skeleton retargeting sits between character animation, heterogeneous motion generation, and
 439 representation learning. Earlier retargeting work established that motion can be shared across bodies
 440 by making skeletal structure explicit, by learning motion spaces intended to be less dependent on
 441 topology, or by conditioning sequence models on skeleton context [Aberman et al., 2020, Lee et al.,
 442 2023, Wang et al., 2024]. The generative setting studied here inherits this ambition, but places it in
 443 sparse heterogeneous libraries where dense source-target correspondences are absent [Gat et al., 2025,
 444 Kim et al., 2025, Li et al., 2023, Chen et al., 2025b]. Our distinction is not that these methods fail
 445 to synthesize plausible target motion. It is that standard evidence can accept a target motion whose
 446 action is correct while the source-conditioned map remains unidentified.

447 Recent work on broader object and animal motion further clarifies why this identification question
 448 matters. Together, these papers move motion synthesis away from one canonical human body and
 449 toward settings in which body identity, morphology, and topology become part of the generative
 450 problem itself [Raab et al., 2024, Lee et al., 2025, Egan et al., 2024, Chen et al., 2025a]. These works
 451 broaden the range of bodies that motion models must handle. The structural question in this paper is
 452 complementary: in sparse heterogeneous data, which evidence selects the motion intention carried
 453 from the source clip rather than a target-side action habit?

Table 2: Truebones zoo regime. The SIF diagnostic is evaluated only on triples with at least three source clips, while the cross-method intersection keeps the triples scored by all evaluated methods.

Quantity	Value	Evidence
Skeletons	70	§1.1
Total clips	616	§1.1
Coarse clusters	10	§1.1
Exact actions	90	§1.1
Skeleton-action cells	6,300 (70×90)	§1.1
Occupied cells	413 (6.6%)	§1.1
Median clips per occupied cell	1	§1.1
SIF eligibility (≥ 3 source clips per triple)	49 triples (171 queries)	§5.1.0
SIF cross-method intersection	37 triples (130 queries)	§5.1.0

The formal argument is closest in spirit to identifiability limits for unsupervised representation learning. Locatello et al. [Locatello et al., 2019] show that unsupervised disentanglement cannot be guaranteed without inductive bias, while the iVAE analysis of Khemakhem et al. [Khemakhem et al., 2020] shows how suitable auxiliary variables can make nonlinear independent components identifiable. Our gauge theorem is narrower than either statement. It does not rule out identification with stronger side information; it shows that per-skeleton marginal matching does not select the relative latent alignment required by a source-preserving retargeting map.

Retrieval and motion-library methods provide a useful foil because they operate directly in motion space. Motion graphs [Kovar et al., 2002] already exploit the idea that a library can preserve physically realized motion snippets, while sparse-correspondence retrieval shows how such libraries can be used when source and target bodies do not share topology [Chen et al., 2025b]. These methods motivate the diagnostic comparison used in the main text: retrieval can characterize what action-level evidence accepts, but by itself it does not identify the source-conditioned retargeting map.

Schrödinger bridge and flow objectives offer another route through the same space of ambiguities. Diffusion Schrödinger bridges [Bortoli et al., 2021] and unpaired neural Schrödinger bridges [Kim et al., 2024] motivate transport between distributions, and flow matching is one way modern retargeting methods instantiate this distributional view [Kim et al., 2025]. In the sparse cross-skeleton setting, distributional transport alone does not remove the relative-gauge ambiguity. This is why the paper separates action-level transport evidence from the source-instance evidence measured by SIF.

B Data Regime and Method Roster

The structural limits studied in the main paper arise in a sparse heterogeneous motion library. Table 2 records the Truebones regime used throughout the experiments. The important fact is not merely that the library is small, but that most skeleton-action cells are empty and most occupied cells contain a single clip, making instance-level correspondence rare even when action labels are available.

The evaluated roster in Table 3 separates published generative models, supervision-matched comparators, the deterministic ANCHOR comparator, and source-blind random references. The public method names in this table are the only names used for result interpretation.

Source-conditioning adaptations. Two rows require additional care because their original public form does not match the SIF input contract. AnyTop is unconditional with respect to a source clip, so the evaluated row adds a self-supervised source-motion encoder with four learned motion queries and injects the resulting latent through cross-attention into the target-skeleton decoder; training remains unpaired self-reconstruction on Truebones. Motion2Motion is a sparse-correspondence retrieval-and-blending procedure rather than a learned generator, so we report both Motion2Motion-Direct, a reimplementation on the 13-channel representation, and Motion2Motion-BVH, the official BVH implementation, using auto-heuristic correspondences and excluding evaluator candidate clips from the target example pool. These details affect only how each row receives source information; SIF is computed by the same source-target-action protocol across rows.

Table 3: Method roster. Train-skeleton counts distinguish transductive evaluation on all skeletons from inductive evaluation with held-out target skeletons.

Method	Family	Train skeletons
AnyTop	source-conditioned diffusion	70
ACE-T	motion-space adversarial	70
ACE-I	motion-space adversarial	60
MoReFlow-T	flow matching	70
MoReFlow-I	flow matching	60
AL-Flow	label-conditional flow, no source motion	60
AL-Flow-Src	label-conditional flow + source motion + skeleton id	60
AL-Flow-Src-G	label-conditional flow + source motion + skeleton graph	60
Motion2Motion-Direct	paired bone regression	60
Motion2Motion-BVH	paired BVH regression	60
DPG-SB-v3	latent Schrödinger bridge	60
ANCHOR	gauge-invariant retrieval (comparator)	60
random-same-cluster	trivial baseline	—
random-same-exact-action	trivial baseline	—

The AnyTop row uses the source-conditioned extension required by SIF; the original generator is unconditional with respect to a source clip. AL-Flow denotes the label-conditional flow family. ANCHOR is a deterministic gauge-invariant retrieval comparator, not a retargeting solution. DPG-SB-v3 is evaluated in its latent-bridge form without decoded-motion penalties (see Appendix K).

C Proofs and Formal Details

C.1 Proof of Theorem 1

Proof. For each skeleton s and motion $x \in \mathcal{X}_s$, the reparameterized encoder-decoder pair satisfies

$$D'_s(E'_s(x)) = D_s(g_s^{-1}(g_s(E_s(x)))) = D_s(E_s(x)),$$

because g_s is a measurable bijection. The per-skeleton reconstruction term $L_s(D_s \circ E_s; P_s)$ is therefore unchanged pointwise.

The latent marginal induced by the reparameterized encoder is $(E'_s)_\# P_s = (g_s \circ E_s)_\# P_s = (g_s)_\# \mu_s$. Since $g \in G_\Omega(\mu)$, the marginal regularizer also remains unchanged:

$$\Omega((g_1)_\# \mu_1, \dots, (g_K)_\# \mu_K) = \Omega(\mu_1, \dots, \mu_K).$$

Combining the unchanged reconstruction terms with the unchanged regularizer gives $L(E', D') = L(E, D)$.

For the retargeting map, define

$$A = \{z : D_b(g_b^{-1}(g_a(z))) \neq D_b(z)\}.$$

By the decoder non-degeneracy hypothesis in Theorem 1, this set has positive μ_a -measure for some pair a, b . Let $B = E_a^{-1}(A) \subseteq \mathcal{X}_a$. Since $(E_a)_\# P_a = \mu_a$, $P_a(B) = \mu_a(A) > 0$. For every $x \in B$,

$$T'_{a \rightarrow b}(x) = D_b(g_b^{-1}(g_a(E_a(x)))) \neq D_b(E_a(x)) = T_{a \rightarrow b}(x),$$

so the two retargeting maps differ on a set of positive P_a -measure. The objective therefore determines the map only up to the relative gauge $g_b^{-1} \circ g_a$. $\square$

Corollary 1.2 (Orthogonal-gauge lower bound under exact Gaussian alignment). *Suppose $\Omega(\mu_1, \dots, \mu_K) = \sum_s \text{KL}(\mu_s \| \mathcal{N}(0, I_d))$, or any rotation-invariant function of the marginals, and each $\mu_s = \mathcal{N}(0, I_d)$. Then G_Ω contains $O(d)^K$, the product of orthogonal groups. The full product has $K d(d-1)/2$ continuous parameters. After quotienting the diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, the relative gauge retains $(K-1)d(d-1)/2$ continuous degrees of freedom.*

511 *Proof.* If $R \in O(d)$ and $z \sim \mathcal{N}(0, I_d)$, then $Rz \sim \mathcal{N}(0, RR^\top) = \mathcal{N}(0, I_d)$. Thus every per-
512 skeleton choice $(R_1, \dots, R_K) \in O(d)^K$ preserves all Gaussian-aligned marginals and belongs to
513 G_Ω whenever Ω is rotation-invariant. Since $O(d)$ has dimension $d(d-1)/2$, the product group
514 has $K d(d-1)/2$ continuous parameters. Quotienting the diagonal subgroup removes one $O(d)$
515 factor, because a common orthogonal transformation applied to all skeletons cancels in the relative
516 transformation $g_b^{-1} \circ g_a$, leaving $(K-1)d(d-1)/2$ relative-gauge parameters. For the setting
517 used in the paper, $K = 70$ and $d = 32$, giving $69 \cdot 32 \cdot 31/2 = 34,224$ relative-gauge degrees of
518 freedom in this orthogonal subgroup. This is a lower bound on relative-gauge degrees of freedom,
519 not by itself a count of distinct decoded retargeting maps; that latter count also depends on decoder
520 non-degeneracy. $\square$

521 C.2 Proof of Proposition 1.1

522 *Proof.* Condition on a source clip x_a and target skeleton b , and write $c = \text{action}(x_a)$. Under
523 the proposition’s sampling rule, the target x_b is drawn uniformly from the finite cell $B(b, c) =$
524 $\{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$, independently of the source instance once the action cell is
525 fixed. For any proposed prediction $u = f(x_a, b)$, the sampling rule gives $x_b \perp x_a \mid (b, c)$, hence
526 $\mathbb{E}[x_b \mid x_a, b] = \mathbb{E}[x_b \mid b, c]$. Therefore

$$\mathbb{E}\|u - x_b\|_2^2 = \|u - \mathbb{E}[x_b \mid b, c]\|_2^2 + \mathbb{E}\|x_b - \mathbb{E}[x_b \mid b, c]\|_2^2.$$

527 The second term is independent of u , so the finite-cell minimizer is unique; in the population version
528 the same equality gives the almost surely unique conditional-mean predictor $u = \mathbb{E}[x_b \mid b, c]$. In the
529 empirical finite-cell case, this conditional expectation is

$$|B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b.$$

530 It depends on the target skeleton and source action, but not on the specific source clip. The Bayes
531 predictor is therefore the target-action cell mean. $\square$

532 C.3 Proof of Corollary 1.1

533 *Proof.* The retrieval rule r depends on the source motion through a feature $\phi(x_a)$ that is computed
534 directly from motion space and is invariant to the latent-gauge transformations in G_Ω . Replacing E_a
535 by $g_a \circ E_a$ therefore changes no argument supplied to r , and the returned library clip is unchanged.
536 Since $r(x_a, b)$ is a single existing target-skeleton clip rather than an average over target-cell motions,
537 the conditional-mean operator in Proposition 1.1 is not the relevant limiting operation. $\square$

538 D Decoder Non-Degeneracy Boundary

539 The gauge theorem requires a decoder that can make at least part of the gauge orbit observable. In
540 a small Kullback-Leibler (KL)-regularized VAE, random gauge rotations and matched-magnitude
541 noise remain near the noise floor in decoded effects, with a gauge-to-noise ratio of 1.11. This is the
542 boundary case in which the decoder nearly absorbs the orbit.

543 A separate AnyTop transformer diffusion hidden-state sweep tests decoder observability, not the
544 $d = 32$ gauge-count model. Across hook layers, reverse timesteps, source motions, and target
545 skeletons, all 72 tested cells, formed by four skeletons, two source clips, three hook layers, and three
546 reverse timesteps, give rotation-induced output divergence above the matched-noise control, with
547 a minimum gauge-to-noise ratio of 1.71. This check does not instantiate the 34,224-dimensional
548 lower-bound calculation. It shows only that decoder observability can hold in a real transformer
549 diffusion motion model, leaving gauge-element selection as the structural problem. Figure 5 and
550 Table 4 summarize these rotation-versus-noise perturbation checks.

551 The cross-seed latent checks in Tables 5 and 6 give a complementary view. Orthogonal Procrustes
552 alignment [Goodall, 1991] explains a substantial fraction of cross-seed latent variation, while effective
553 rank separates rich latent variation from low-dimensional collapse. These checks defend the boundary
554 of the theory: decoder observability can hold even when the objective does not select the relative
555 gauge. Figures 6, 7, and 8 visualize the same distinction between latent alignment, representation
556 richness, and source-instance fidelity.

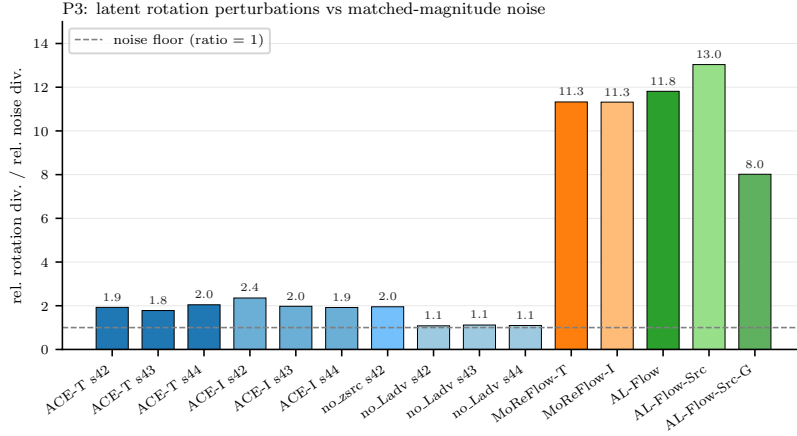


Figure 5: Decoder non-degeneracy check. Rotation-induced changes are compared with matched-magnitude noise in the latent space; ratios above one indicate that the decoder distinguishes at least some directions on the gauge orbit.

Table 4: Rotation versus noise perturbations on the SIF-intersection queries. The ACE ablation with the motion-space adversarial loss removed is the boundary case near the noise floor, while the other evaluated methods make the orbit visible after decoding.

Method	Seed	rel. div. rotation	rel. div. noise	Ratio	Verdict
ACE-T (3 seeds)	42	0.240	0.213	1.93	non-degenerate
	43	0.219	0.183	1.78	non-degenerate
	44	0.209	0.171	2.04	non-degenerate
ACE-I (3 seeds)	42	0.121	0.089	2.35	non-degenerate
	43	0.128	0.071	1.98	non-degenerate
	44	0.141	0.087	1.92	non-degenerate
No z_{src} ablation	42	0.117	0.083	1.95	non-degenerate
No L_{adv} ablation (3 seeds)	42	1.113	1.154	1.08	at noise floor
	43	0.810	0.849	1.12	at noise floor
	44	0.699	0.734	1.10	at noise floor
MoReFlow-T	–	0.113	0.017	11.33	non-degenerate (strong)
MoReFlow-I	–	0.107	0.016	11.32	non-degenerate (strong)
AL-Flow	–	0.052	0.007	11.81	non-degenerate (strong)
AL-Flow-Src	–	0.065	0.009	13.04	non-degenerate (strong)
AL-Flow-Src-G	–	0.048	0.008	8.02	non-degenerate (strong)

Each row reports the mean over 130 SIF-intersection queries with 20 random $O(d)$ rotations + 20 matched-magnitude noise samples per query. The no- L_{adv} ablation is statistically indistinguishable from the noise floor (Wilcoxon $p > 0.4$ for all three seeds); every other method exceeds the noise floor at $p < 10^{-12}$.

557 E Source-Instance Fidelity Robustness

558 SIF is a diagnostic for whether source-side geometry remains visible after the target skeleton and
559 action are fixed. The main text reports the qualitative pattern; Tables 7 and 8 give the max-support
560 and intersection estimates behind that pattern.

561 The robustness checks ask whether the same ordering survives dependence across source skeletons,
562 target skeletons, and actions. Table 9 reports clustered bootstrap intervals, and Table 10 reports paired
563 comparisons on the same 37 triples. Figure 9 visualizes the clustered intervals. The conclusion is the
564 same as in the main text: ACE-I and ACE-T are the robust nontrivial positives on the intersection,
565 random-same-exact-action is a label-only positive, and MoReFlow-T remains borderline.

Table 5: Orthogonal Procrustes alignment across seeds. The table reports how much latent variation can be explained by an $O(d)$ alignment across the SIF-intersection queries.

Group	Seed pair	Frac. $O(d)$ -explained	Row-cosine
ACE-T	s42 vs s43	0.522	0.893
ACE-T	s42 vs s44	0.703	0.900
ACE-T	s43 vs s44	0.585	0.905
ACE-I	s42 vs s43	0.689	0.935
ACE-I	s42 vs s44	0.712	0.928
ACE-I	s43 vs s44	0.775	0.941
no_L_adv	s42 vs s43	0.482	0.866
no_L_adv	s42 vs s44	0.428	0.847
no_L_adv	s43 vs s44	0.753	0.881

Frac. $O(d)$ -explained is the fraction of cross-seed latent variance accounted for by the optimal $O(d)$ Procrustes rotation. *Row-cosine* is the mean per-row cosine similarity after alignment. Both are computed on per-target-skeleton latents across 130 SIF-intersection queries, aggregated over 21 target skeletons.

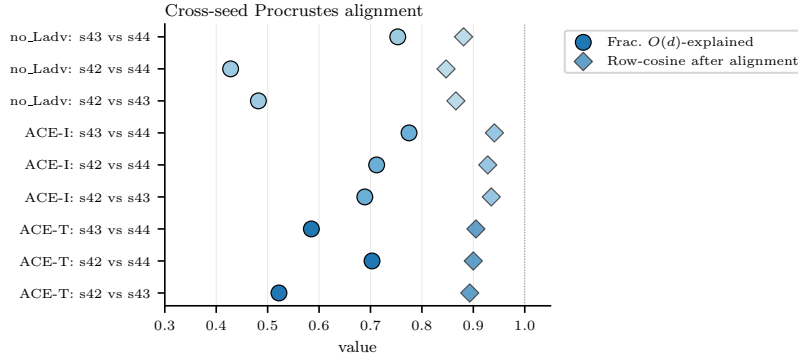


Figure 6: Cross-seed Procrustes alignment. High explained fractions indicate that latent spaces differ partly by approximately orthogonal transformations, matching the relative-gauge mechanism in Theorem 1.

566 The eligibility rule also matters for interpretation. Table 11 and Figure 10 show that SIF triples have
567 at least three source clips by construction, but only a few triples provide more than three. Figure 11
568 checks that the main pattern is not a monotone artifact of the diversity ratio R .

569 Latent SIF (L-SIF) repeats the SIF computation before decoding, when a method exposes a latent
570 representation on the relevant queries. Table 12 and Figure 12 show that latent preservation and
571 decoded SIF can decouple. This motivates the failure-mode taxonomy used in the qualitative
572 discussion: a method may collapse in latent space, collapse after decoding, or recover output variation
573 without proving source memorization.

574 F Synthetic Calibration

575 The synthetic 2×2 environment calibrates SIF in a setting where the source-conditioned transport
576 is known. Figure 13 visualizes the three reference regimes used in Section 5: the oracle transport
577 lies near one, while two source-blind references lie near zero. Table 13 records the corresponding
578 numerical calibration.

579 The same environment also isolates conditional-mean degeneration. The cell-paired squared-error
580 predictor receives random target samples from the correct target skeleton and action cell, but not
581 instance-level pairs. As Table 14 shows, increasing the number of random cell-pairs does not recover
582 the oracle within-cell variance.

Table 6: Effective-rank diagnostic on latent distributions. Higher effective rank indicates richer latent variation, but rank alone does not imply that the source-conditioned map has been identified.

Method	Sample shape	Spectral flatness	Effective rank / 256
no_L_adv s42	$3,376 \times 256$	0.0000	1.29
ACE-T s42	$3,376 \times 256$	0.0000	2.13
ACE-I s42	$3,376 \times 256$	0.0019	3.00
AL-Flow	$1,040 \times 256$	0.0101	5.84
MoReFlow-I	$3,823 \times 256$	0.0593	12.84
MoReFlow-T	$3,823 \times 256$	0.0631	14.99
AnyTop encoder z	$13,000 \times 256$	0.1049	41.41

Effective rank computed on the per-method latent distribution over the 130 SIF-intersection queries.
AnyTop’s encoder hidden state is an order of magnitude richer than the Stage-A bottleneck latents.

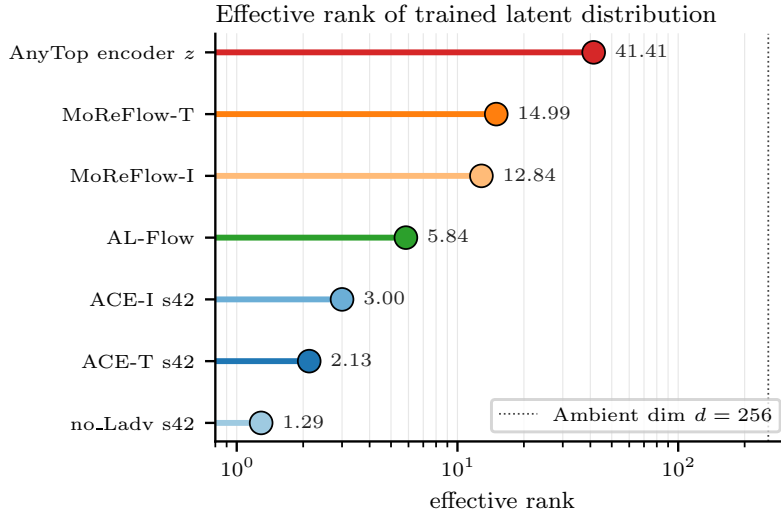


Figure 7: Effective rank across latent representations. The diagnostic separates representation richness from source-instance fidelity.

583 G Deterministic Comparator and Motion Descriptors

584 ANCHOR is included to make the diagnostic boundary explicit. It is not a generator and does not
585 synthesize a new target motion. For a query consisting of a source clip, a target skeleton, and an
586 action request, ANCHOR first predicts a coarse source action cluster using a random-forest classifier
587 [Breiman, 2001] trained on motion-space descriptors from the training skeletons. It then searches the
588 target skeleton’s motion library and returns one existing clip.

589 The motion descriptor, denoted $Q(x)$, is computed from joint-position trajectories rather than from a
590 learned latent representation. It summarizes center-of-mass displacement and speed, heading velocity,
591 contact timing, cadence, dominant limb usage, and global clip statistics. The candidate pool is the
592 union of all target-skeleton clips in the predicted cluster and the ten clips with largest Q -descriptor
593 similarity to the source. Candidate clips are scored by cluster agreement, descriptor similarity,
594 and exact-action agreement, using the fixed comparator weights $(w_{\text{cluster}}, w_Q, w_{\text{action}}) = (1, 2, 3)$
595 recorded in the method roster.

596 This construction is why ANCHOR is gauge-invariant in the sense of Corollary 1.1: changing a
597 learned latent parametrization cannot change a score computed directly from center-of-mass, heading,
598 contact, cadence, and limb-usage measurements on the joint trajectories. At the same time, because
599 ANCHOR returns an existing library clip, it does not claim to identify the source-conditioned
600 retargeting map. Its purpose is to mark a diagnostic boundary, showing that action recovery can

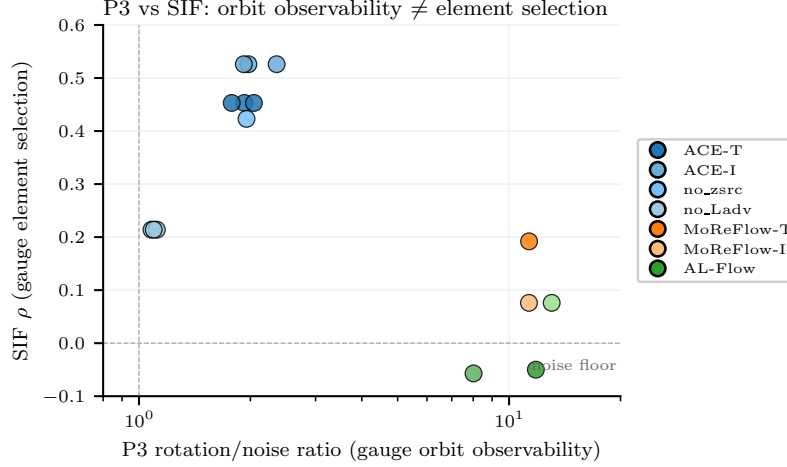


Figure 8: Decoder non-degeneracy and SIF are distinct. A method may expose latent perturbations after decoding while still lying near the source-blind floor under SIF.

Table 7: Max-support SIF results. The table reports each method on the largest set of eligible triples it can score, together with the corresponding 37-triple intersection interval.

Method	ρ_{SIF} (max)	n	R	95% CI on intersection
ACE-I	0.484	49	0.11	0.526 [0.349, 0.677]
ACE-T	0.385	49	14.94	0.453 [0.293, 0.604]
MoReFlow-T	0.203	49	0.56	0.192 [0.006, 0.378]
MoReFlow-I	0.137	49	0.56	0.076 [-0.150, 0.270]
AL-Flow-Src	0.070	38	0.22	0.076 [-0.110, 0.260]
Motion2Motion-Direct	0.053	49	2.42	0.131 [-0.050, 0.300]
ANCHOR	0.014	49	0.17	0.040 [-0.070, 0.170]
AL-Flow-Src-G	-0.028	49	0.18	-0.057 [-0.230, 0.150]
AnyTop	-0.033	49	3.01	-0.010 [-0.180, 0.170]
DPG-SB-v3	-0.058	49	0.58	-0.019 [-0.220, 0.180]
AL-Flow	-0.066	45	0.16	-0.050 [-0.230, 0.130]
Motion2Motion-BVH	-0.085	49	1.31	-0.067 [-0.240, 0.090]
Random baselines:				
random-same-exact-action	0.096	48	0.36	0.186 [0.040, 0.330]
random-same-cluster	0.073	48	0.83	0.104 [-0.090, 0.280]

Identity reference: $\rho_{\text{SIF}} = +1.000$. Cell-mean reference: $\rho_{\text{SIF}} = 0.000$. The ρ column reports each method’s maximal-support estimate (n in the third column); the 95% CI column is from the cross-method 37-triple intersection bootstrap. **Bold** marks methods whose intersection lower CI exceeds +0.01 (*robust positives*); MoReFlow-T at +0.006 is treated as borderline and is not bolded (see `tab_sif_intersect`).

601 remain high at the source-blind floor. Table 15 records the action-cluster predictor, and Figure 14
602 shows how ANCHOR matches vary across action clusters.

603 H Action-Level AUC and Label Recovery

604 The standard cross-skeleton contrastive retrieval AUC is an action-level ranking test: it asks whether
605 a generated target motion ranks positives above negatives in a target library. Section 5 shows that
606 this test can accept label-only behavior on the covered queries. The appendix gives the broader
607 enumeration and comparison tables behind that statement. Table 16 and Figure 15 report the full
608 enumeration, while Figure 16, Table 17, Table 18, and Figure 17 collect the restricted-query and
609 AUC-band controls.

Table 8: SIF on the 37-triple cross-method intersection. This is the common support used for clustered intervals and paired comparisons.

Method	ρ_{SIF} (95% CI)	R (95% CI)	Status
ACE-I	0.526 [0.349, 0.677]	0.10 [0.04, 0.18]	robust
ACE-T	0.453 [0.293, 0.604]	4.59 [0.36, 11.06]	robust
MoReFlow-T	0.192 [0.006, 0.378]	0.53 [0.34, 0.77]	borderline
random-same-exact-action	0.186 [0.040, 0.330]	0.32 [0.15, 0.57]	robust
Motion2Motion-Direct	0.131 [-0.050, 0.300]	2.05 [1.50, 2.71]	
random-same-cluster	0.104 [-0.090, 0.280]	0.74 [0.45, 1.11]	
MoReFlow-I	0.076 [-0.150, 0.270]	0.57 [0.33, 0.90]	
AL-Flow-Src	0.076 [-0.110, 0.260]	0.23 [0.14, 0.34]	
ANCHOR	0.040 [-0.070, 0.170]	0.20 [0.06, 0.40]	
AnyTop	-0.010 [-0.180, 0.170]	2.98 [1.78, 4.62]	
DPG-SB-v3	-0.019 [-0.220, 0.180]	0.58 [0.33, 0.92]	
AL-Flow	-0.050 [-0.230, 0.130]	0.10 [0.06, 0.15]	
AL-Flow-Src-G	-0.057 [-0.230, 0.150]	0.19 [0.11, 0.30]	
Motion2Motion-BVH	-0.067 [-0.240, 0.090]	1.32 [0.91, 1.81]	

$n = 37$ cross-method intersection. Identity reference: $\rho = +1.000$. Cell-mean reference: $\rho = 0.000$. **Status** marks the lower-CI position relative to zero: *robust* if lower CI $> +0.01$; *borderline* if lower CI is in $(0, +0.01]$ (MoReFlow-T at $+0.006$).

Table 9: Clustered bootstrap intervals for SIF. Clustering by source skeleton, target skeleton, or action tests whether the positive conclusions are driven by a narrow subset of triples.

Method	iid	cluster: source-skel	cluster: target-skel	cluster: action
ACE-I	0.526 [0.351, 0.681]	0.526 [0.323, 0.709]	0.526 [0.365, 0.671]	0.526 [0.242, 0.629]
ACE-T	0.453 [0.293, 0.602]	0.453 [0.284, 0.618]	0.453 [0.274, 0.605]	0.453 [-0.009, 0.619]
MoReFlow-T	0.192 [-0.003, 0.380]	0.192 [-0.023, 0.386]	0.192 [-0.031, 0.405]	0.192 [-0.039, 0.357]
random-same-exact-action	0.185 [0.046, 0.326]	0.185 [0.069, 0.304]	0.185 [0.027, 0.354]	0.185 [0.000, 0.272]
Motion2Motion-Direct	0.131 [-0.061, 0.317]	0.131 [-0.098, 0.385]	0.131 [-0.075, 0.307]	0.131 [-0.720, 0.265]
random-same-cluster	0.104 [-0.095, 0.298]	0.104 [-0.057, 0.279]	0.104 [-0.115, 0.315]	0.104 [-0.272, 0.189]
AL-Flow-Src	0.076 [-0.108, 0.264]	0.076 [-0.109, 0.269]	0.076 [-0.112, 0.248]	0.076 [-0.222, 0.169]
MoReFlow-I	0.076 [-0.132, 0.279]	0.076 [-0.132, 0.271]	0.076 [-0.167, 0.319]	0.076 [-0.108, 0.474]
ANCHOR	0.040 [-0.068, 0.157]	0.040 [-0.078, 0.176]	0.040 [-0.068, 0.167]	0.040 [0.000, 0.046]
AnyTop	-0.010 [-0.185, 0.166]	-0.010 [-0.191, 0.166]	-0.010 [-0.207, 0.185]	-0.010 [-0.253, 0.142]
DPG-SB-v3	-0.019 [-0.215, 0.178]	-0.019 [-0.208, 0.188]	-0.019 [-0.244, 0.228]	-0.019 [-0.487, 0.045]
AL-Flow	-0.050 [-0.231, 0.137]	-0.050 [-0.187, 0.106]	-0.050 [-0.259, 0.161]	-0.050 [-0.239, 0.183]
AL-Flow-Src-G	-0.057 [-0.239, 0.130]	-0.057 [-0.207, 0.087]	-0.057 [-0.261, 0.128]	-0.057 [-0.523, 0.022]
Motion2Motion-BVH	-0.067 [-0.237, 0.110]	-0.067 [-0.247, 0.116]	-0.067 [-0.221, 0.089]	-0.067 [-0.320, 0.142]

Bold lower CI excludes zero. Bootstrap $N = 10,000$, seed = 42. ACE-I is the only method whose lower CI excludes zero under all four bootstrap schemes; ACE-T and random-same-exact-action survive source-skel and target-skel clusterings but action-clustered drops their lower CI to or below zero due to action concentration in the 37-triple intersection.

610 I ACE Controls

611 ACE is the characterized positive case in this study because its motion-space constraint narrows the
612 observational equivalence class more effectively than the other evaluated objectives. The ablation
613 in Table 19 and Figure 18 supports this interpretation: removing the motion-space adversarial term
614 reduces SIF across three seeds, while the full ACE-T and ACE-I models remain the robust nontrivial
615 positives on the intersection.

616 The nearest-neighbor leakage check in Table 20 addresses a different explanation: that ACE could be
617 lifting SIF by memorizing source clips. Most ACE outputs are closer to a target-pool training clip
618 than to the corresponding source clip in the reported four-dimensional kinematic feature space, which
619 weakens source memorization as the explanation for the SIF lift.

620 J Qualitative Evidence and Supplementary Videos

621 The qualitative figures are not used as quantitative proof, but they make the SIF failure modes visible.
622 The appendix provides three full-grid still figures that correspond to the supplementary videos. Each

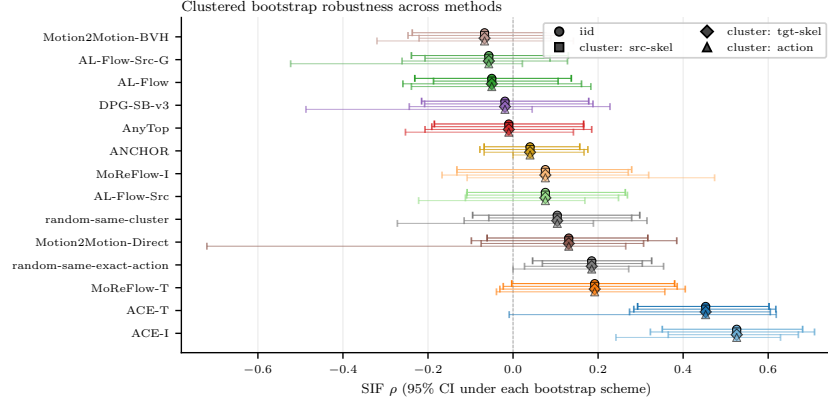


Figure 9: Clustered confidence intervals for SIF. The figure visualizes the dependence-aware intervals reported in Table 9.

Table 10: Paired SIF comparisons on the 37-triple intersection. Paired tests compare methods on identical triples rather than across different supports.

Pairing	Mean $\Delta\rho$	Wilcoxon p	Bootstrap 95% CI on $\Delta\rho$	Excludes zero
ACE-I vs random-same-exact-action	+0.340	4.35×10^{-3}	[+0.127, +0.541]	yes
ACE-T vs random-same-exact-action	+0.268	7.94×10^{-3}	[+0.047, +0.477]	yes
ACE-I vs MoReFlow-T (next non-trivial)	+0.334	7.21×10^{-3}	[+0.089, +0.582]	yes
ACE-T vs MoReFlow-T	+0.262	5.27×10^{-2}	[-0.004, +0.533]	no (borderline)
ACE-T vs no_L_adv (3-seed mean)	+0.153	6.10×10^{-3}	—	yes
ACE-I vs no_L_adv (3-seed mean)	+0.312	1.70×10^{-3}	—	yes

All paired tests on the 37-triple intersection ($n = 37$). ACE-I separates from both the strongest source-blind reference (random-same-exact-action) and the next non-trivial generative method (MoReFlow-T); ACE-T separates from random-same-exact-action but is borderline against MoReFlow-T.

grid fixes a source-target triple and compares the evaluated methods across three source clips, so rows that move in near lockstep indicate source-blind behavior while rows with source-dependent variation indicate partial source conditioning. Figures 19, 20, 21, 22, and 23 provide the still-image views corresponding to this qualitative protocol.

The supplementary material contains nine companion videos: two main-body videos for the panorama and failure-mode figures, and seven appendix videos for additional source-target triples. The videos use the same projection rule as the still figures and synchronize playback phase across methods, making source-blind lockstep behavior and source-conditioned variation easier to see.

K Latent Schrödinger Bridge Comparator Scope

The evaluated latent Schrödinger bridge comparator, reported under the public method name DPG-SB-v3 in the roster, uses retrieval-initialized noise, a shared conditioned bridge, a target-side latent constraint, and a latent-space cycle term. The decoded-motion bone, contact, and smoothness penalties were not included in this evaluation. This caveat is important for interpretation: DPG-SB-v3 tests where the evaluated latent bridge variant lies relative to the source-blind floor, not an exhaustive set of motion-space penalties for bridge objectives.

L Potential Societal Impacts

Cross-skeleton motion generation may broaden access to motion content in animation, education, assistive design, and scientific visualization by reducing the need to record every action separately for every body. In these settings, a source-preserving model could help creators transfer motion intention across characters, support low-resource motion libraries, and make comparative studies of movement

Table 11: Pair counts per SIF triple. The minimum and median values are fixed by the eligibility rule; the few larger cells determine how much within-cell geometry is available.

Method	Min pairs	Median pairs	Max pairs	n triples
ACE-I	3	3	5	49
ACE-T	3	3	5	49
AL-Flow	3	3	5	45
AL-Flow-Src	3	3	5	38
AL-Flow-Src-G	3	3	5	49
ANCHOR	3	3	5	49
AnyTop	3	3	5	49
DPG-SB-v3	3	3	5	49
Motion2Motion-BVH	3	3	5	49
Motion2Motion-Direct	3	3	5	49
MoReFlow-I	3	3	5	49
MoReFlow-T	3	3	5	49
random-same-exact-action	3	3	5	48
random-same-cluster	3	3	5	48

Min and median are uniformly 3 because the SIF benchmark requires ≥ 3 source clips per triple by design; max reflects the few triples in Truebones with 4 or 5 source clips. Variation across methods comes from the few queries that some baselines fail to score.

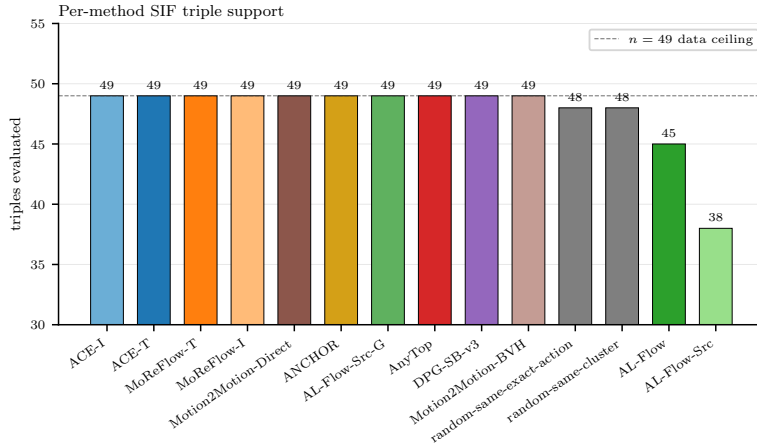


Figure 10: Pair-count distribution for SIF-eligible triples.

643 easier. This paper contributes a diagnostic and theoretical account of when such transfer claims are
644 not identified by standard evidence, rather than a deployed generator.

645 The same capability also carries risks. More reliable motion transfer could make synthetic embodied
646 media easier to produce, including misleading performances attributed to people or characters who
647 did not enact them, and action-level evaluation may overstate reliability when the source-conditioned
648 map is not actually preserved. Biases in the source and target motion libraries may also be propagated
649 across bodies, especially when some anatomies, actions, or movement styles are sparsely represented.
650 Our analysis supports a conservative evaluation stance: cross-skeleton methods should report source-
651 instance diagnostics alongside action-level scores, disclose the motion domains and licenses on which
652 they are trained, and avoid claims of retargeting fidelity when only target-action recovery has been
653 established.

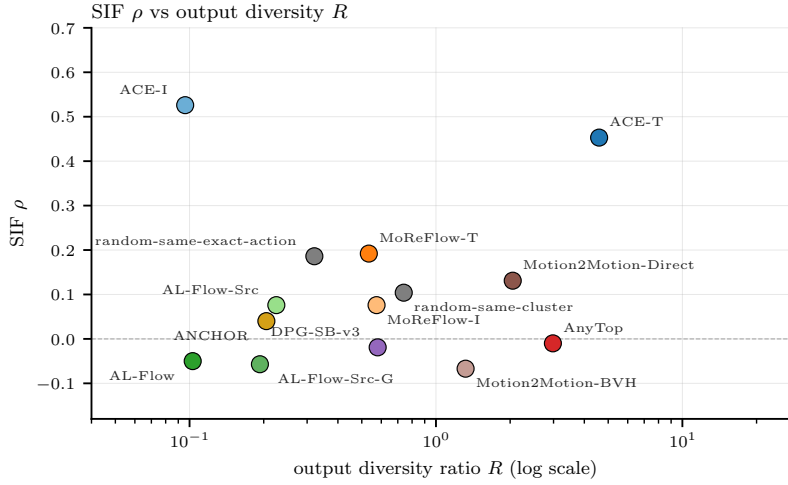


Figure 11: SIF versus output diversity ratio. High output diversity is not sufficient for source-instance preservation, because the geometry must align with the source-side geometry after the target skeleton and action are fixed.

Table 12: SIF and L-SIF failure-mode taxonomy. L-SIF is a latent-space diagnostic and is therefore reported only where the representation is comparable across queries.

Failure mode	Method	SIF ρ	L-SIF ρ
Source-conditional	MoReFlow-T	0.192	0.165
Output-only recovery	ACE-I (3 seeds avg)	0.503	0.083
	ACE-T (3 seeds avg)	0.408	0.088
	no_z_src	0.423	0.107
Mixed (both modest)	no_L_adv (3 seeds avg)	0.214	0.055
	AL-Flow-Src	0.076	0.090
Decoder collapse	MoReFlow-I	0.076	0.132
	AL-Flow-Src-G	-0.057	0.109
	AnyTop	-0.010	0.141
Latent collapse	AL-Flow	-0.050	-0.001

Cross-method Pearson correlation between SIF and L-SIF: $r = -0.003$, $p = 0.99$ across $n = 16$ method runs. Per-method L-SIF confidence intervals individually cross zero, so the four-mode classification is observational across method families rather than a per-method significance claim.

Table 13: Synthetic SIF calibration. The oracle lies near perfect source-instance preservation, while source-blind references remain near zero.

Configuration	ρ_{SIF} (95% CI)	R
Oracle (output = $T^*(x_a)$)	0.9957 [0.9714, 1.0000]	1.16
Random clip from cell	-0.0002 [-0.0870, 0.1382]	19.60
Random Gaussian noise	-0.0009 [-0.1225, 0.1286]	25.70

Synthetic 2×2 paired-dense setting, $n = 336$ pairs. The oracle saturates SIF; both source-blind references sit at zero.

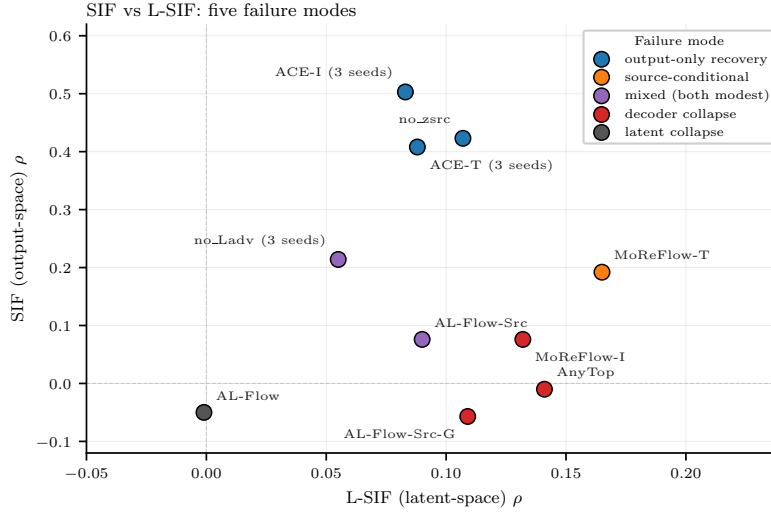


Figure 12: SIF and L-SIF decoupling. Source-instance fidelity in decoded motion need not coincide with a positive latent-space correlation.

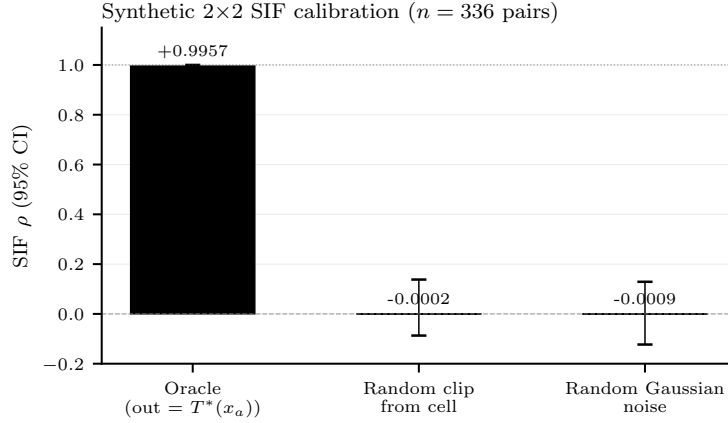


Figure 13: Synthetic 2×2 SIF calibration. The oracle uses the known source-conditioned transport $T^*(x_a)$; the random-clip and Gaussian references deliberately ignore the source instance.

Table 14: Conditional-mean degeneration in the controlled 2×2 setting. The predicted variance remains a small fraction of oracle variance even when the number of random target-cell pairs increases.

M pairs / cell	$\sigma_{\text{pred}}^2 / \sigma_{\text{oracle}}^2$	$\text{MSE} / \sigma_{\text{oracle}}^2$	n_{seeds}
1	0.249 ± 0.047	2.32 ± 2.40	3
2	0.185 ± 0.028	0.99 ± 0.10	3
4	0.175 ± 0.031	0.93 ± 0.02	3
8	0.162 ± 0.011	0.93 ± 0.03	3
16	0.188 ± 0.018	0.92 ± 0.03	3
32	0.158 ± 0.035	0.95 ± 0.09	3
50	0.133 ± 0.023	1.00 ± 0.06	3

Across all M and 3 seeds, the variance ratio $\sigma_{\text{pred}}^2 / \sigma_{\text{oracle}}^2$ stays at 0.13–0.25 even at $M = 50$ pairs per cell, confirming prop: cm predicted cell-mean compression.

Table 15: ANCHOR action-cluster predictor. The held-out row shows that the cluster prediction itself is imperfect, which is why ANCHOR is interpreted as a comparator rather than an oracle.

Domain	Cluster classifier accuracy	n correct / total
In-domain (train skeletons)	99.8%	—
Held-out (reserved cross-skeleton)	61.5%	59 / 96

Replacing the filename cluster with the predicted cluster pulls Procrustes performance below random on exact-tier and on the B1 hand-curated subset. The in-domain row reports only the percentage from the original training log; the corresponding numerator/denominator was not preserved and is therefore listed as *not available*.

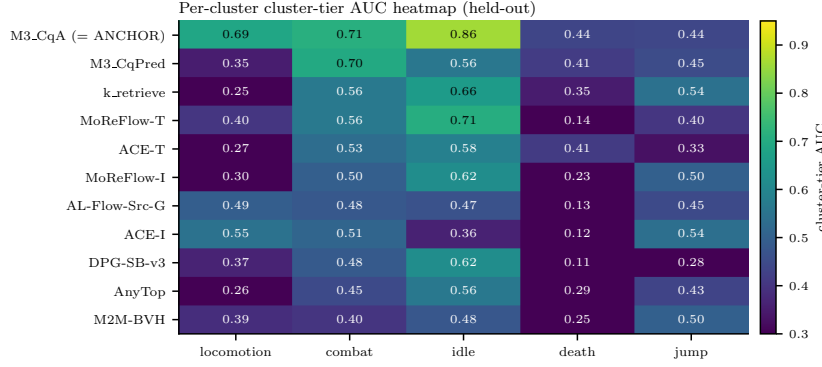


Figure 14: Per-cluster ANCHOR matches. The comparator’s behavior varies across action clusters, reinforcing that it characterizes a floor rather than a universal retargeting rule.

Table 16: Full ANCHOR enumeration across cluster-eligible triples. The held-out subsets show that strong action-level ranking can arise from target-library label structure without identifying a source-conditioned map.

Subset	n triples (cluster-eligible)	Cluster-tier AUC (95% CI)
All	30,029	0.824 [0.820, 0.829]
In-distribution (same train pair)	21,838	0.845 [0.841, 0.849]
Mixed (one held-out skeleton)	7,622	0.774 [0.762, 0.787]
Held-out (both cross-skeleton)	569	0.711 [0.669, 0.758]

ANCHOR on the 30,497-pair full enumeration. Cluster-eligibility filter removes triples whose target skeleton has no in-cluster positives. The mixed subset is the load-bearing held-out generalisation case.

Table 17: Action-level AUC band for generated variants and supervision-matched comparators. These values are not interpreted as source-preserving transfer; they show how much action-level evidence can be explained without SIF.

Variant	Conditioning channels	Cluster AUC	Band
AnyTop	self-supervised source-conditioned diffusion	0.466	inside
Precursor	action-conditioned generation	0.484	inside
Precursor	latent-bridge generator	0.484	inside
DPG-SB-v3	latent bridge objective; no decoded-motion penalties	0.448	inside
AL-Flow	cluster + exact-action labels	0.538	inside
AL-Flow-Src	AL-Flow + source motion + skel. id	0.542	inside
AL-Flow-Src-G	AL-Flow + source motion + skel. graph	0.470	inside

Support-qualified band is [0.39, 0.59] for the seven listed configurations. The two precursor rows are reported only for the action-level AUC-band analysis and are not part of the 14-method SIF roster. The wider [0.37, 0.55] strict-support band covers all 13 evaluated configurations (see `tab_master_auc` for the full 14-method comparison).

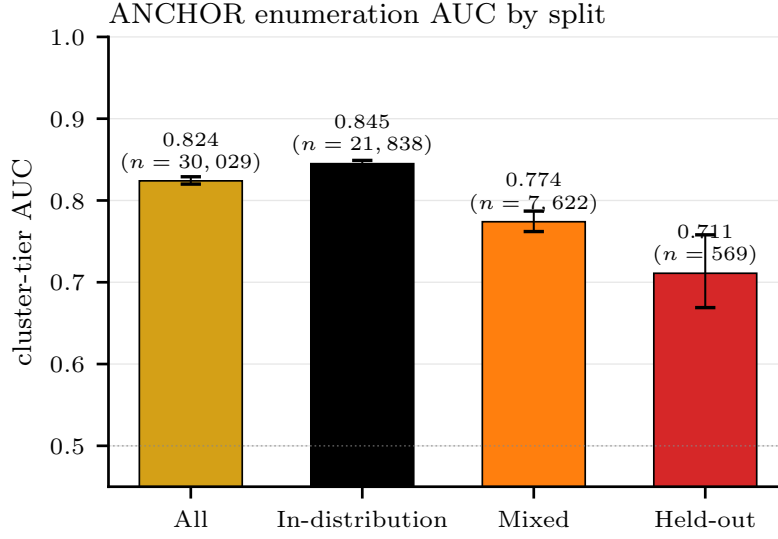


Figure 15: Enumeration breakdown for ANCHOR. The large enumeration separates in-distribution, mixed, and held-out skeleton regimes.

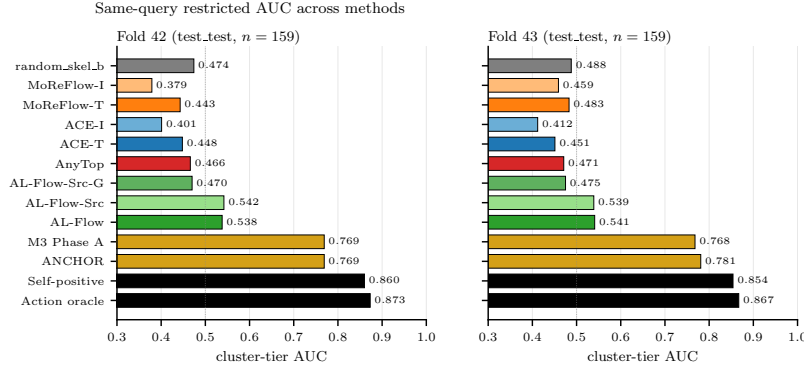


Figure 16: Same-query restricted AUC comparison. Restricting the comparison to shared queries keeps the contrast focused on the action-level ranking signal.

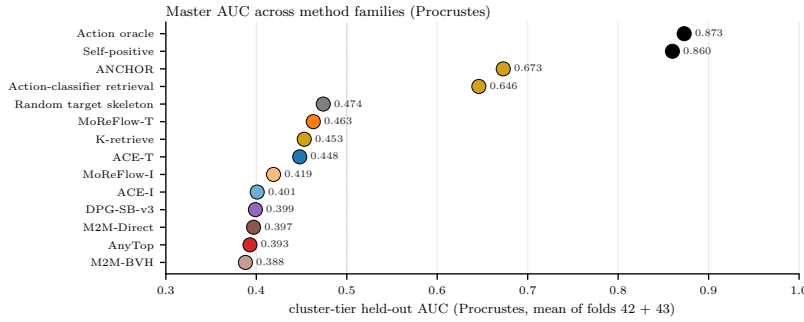


Figure 17: Action-level AUC band. Many methods occupy a narrow action-ranking band, which motivates SIF as the missing source-instance axis.

Table 18: Master action-level AUC comparison on held-out queries. The table reports cluster-tier AUC under three distances and separates library-based references from generative methods.

Method	Cluster-tier held-out AUC (average of folds 42 and 43)		
	Procrustes	Z-DTW	Q-comp
Action oracle	0.873	0.846	0.894
Self-positive reference	0.860	0.847	0.909
ANCHOR	0.673	0.707	0.718
Cluster-Classifer Retrieval	0.646	0.669	0.693
Q-Retrieval (Q-feature only)	0.453	0.523	0.597
MoReFlow-T	0.463	0.485	0.510
MoReFlow-I	0.419	0.426	0.524
ACE-T	0.448	0.451	0.529
ACE-I	0.401	0.469	0.550
Random target skeleton (null)	0.474	0.510	0.530

Motion2Motion-BVH has Procrustes only and is reported separately. MoReFlow-T and MoReFlow-I Procrustes values are the per-fold-corrected averages from EVIDENCE DOSSIER § 9; an earlier internal report had stale values 0.443 and 0.379.

Table 19: ACE ablation controls. Removing the motion-space adversarial loss lowers SIF, while the feature-loss ablation diverges and therefore yields no usable checkpoint.

Variant	Seed	ρ_{SIF}	95% CI	R	Status
ACE-T (full model)	42	0.453	[0.293, 0.604]	4.59	reference
ACE-I (full model)	42	0.526	[0.349, 0.677]	0.10	reference
No z_{src} input	42	0.423	[0.235, 0.581]	0.05	trained, 50k steps
No L_{feat}	—	—	—	—	training diverged (NaN at step 4,600)
No L_{adv}	42	0.175	[−0.028, 0.373]	1.43	trained, 50k steps
No L_{adv}	43	0.191	[0.001, 0.377]	0.59	trained, 50k steps
No L_{adv}	44	0.276	[0.069, 0.465]	0.49	trained, 50k steps
No L_{adv} (mean)	3 seeds	0.214	—	0.84	$\sigma = 0.054$

Three independent retrains for the no- L_{adv} ablation. Removing the motion-space adversarial loss collapses ACE’s source-instance fidelity across all three seeds (paired Wilcoxon $p < 0.01$ vs both ACE-T and ACE-I baselines). The L_{feat} ablation diverges during training and yields no usable checkpoint.

Table 20: ACE nearest-neighbor leakage check. Distances are computed in a kinematic feature space; the majority of ACE outputs are closer to the target pool than to the source clip.

Method	Queries	Closer-to-target-pool	Mean d_{src}	Mean d_{pool}	Ratio
ACE-I	130	103 (79.2%)	3.449	2.215	1.56
ACE-T	130	99 (76.2%)	3.697	2.650	1.39

Distance computed in a four-dimensional kinematic feature space (centre-of-mass displacement, centre-of-mass variance, mean velocity, foot-contact density). 76–79% of ACE outputs are closer to a target-pool training clip than to the corresponding source clip, weakening source-memorisation as an explanation of ACE’s SIF lift.

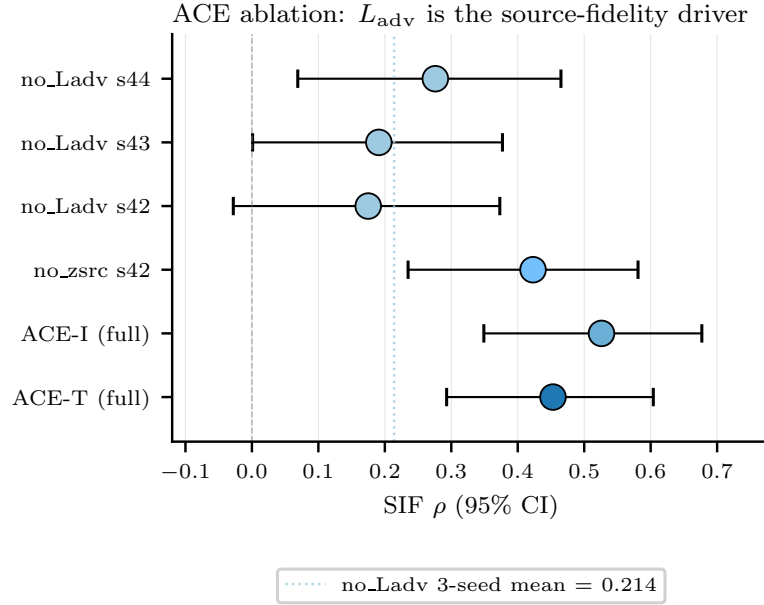


Figure 18: ACE ablation effect on SIF. The motion-space adversarial constraint is the relevant positive mechanism in the evaluated ACE family.

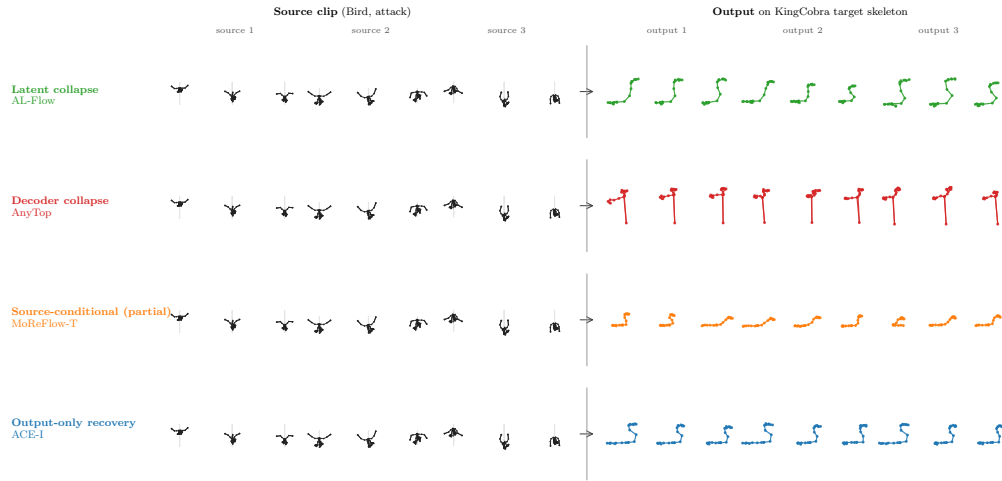


Figure 19: A source-clip substitution view on one Bird $\rightarrow$ KingCobra attack triple. The first three columns, labeled source 1 to source 3, are different Bird attack clips, while the target skeleton and action are fixed; the three columns on the right, labeled output 1 to output 3, show the corresponding KingCobra motions generated by each row. Each row is labeled by its source-instance failure mode: latent collapse for AL-Flow, where source identity is lost before decoding; decoder collapse for AnyTop, where latent variation is not expressed in the output; partial source-conditionality for MoReFlow-T; and output-only recovery for ACE-I, the clearest output-level source dependence among the displayed rows.

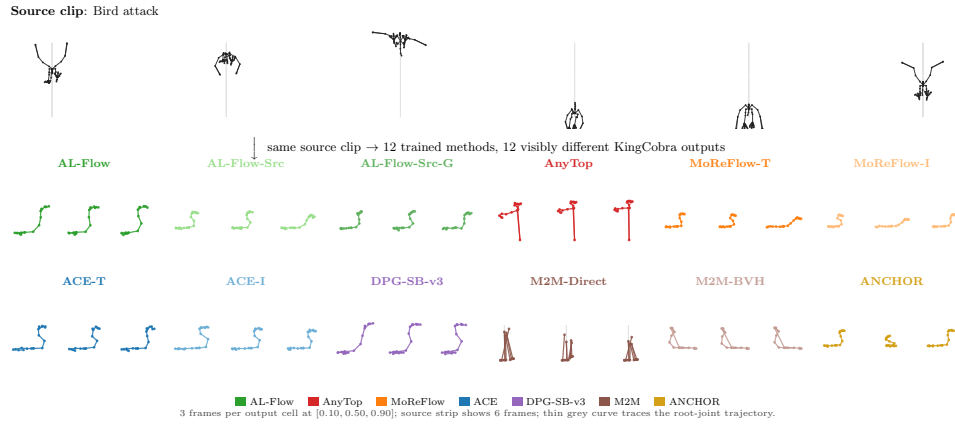


Figure 20: Qualitative method panorama. The figure shows one source-target query across the evaluated methods, emphasizing that action-consistent outputs can differ substantially while the source-conditioned map remains unidentified.



Figure 21: Full qualitative grid A. Three source clips are compared against target outputs from the evaluated methods on a fixed source-target-action triple.

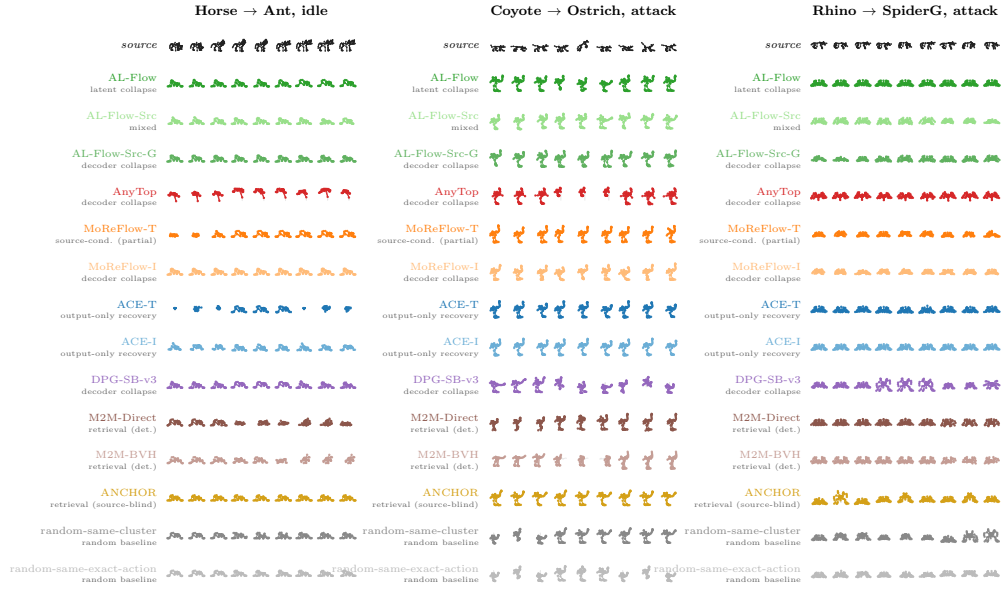


Figure 22: Full qualitative grid B. The same layout is used to reveal whether output variation follows the source clips after target skeleton and action are fixed.

Hippopotamus → Lion, attack



Figure 23: Full qualitative grid C. The grid complements Figure 19 by showing all evaluated methods on an additional held-out triple.

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